SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Short Answer

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – Short Answer
Weightage:	30 % of CIA	**Total Marks:**	100 Marks
CO1 Statement:	*Analyze the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
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Section / Batch:		**Date:**	

Code of Conduct

I B.Janani (192321100) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: B.Janani

Instructions

- This test contains 50 questions, Total: 100 Marks.
- Duration: 1 hour.

70/100

Questions

Q1. A telecom operator wins a spectrum auction for 45 MHz. Company policy sets aside 25 kHz for every simplex channel. The planning team wants to know how many full-duplex voice channels this spectrum can support.

Q2. A regional carrier runs its network on 560 total voice channels, shared across a 7-cell frequency reuse cluster. During capacity planning, an engineer needs to know how many channels a single cell will typically be allocated.

Q3. A GSM operator sets carrier spacing at 200 kHz and rolls out 100 carriers, each carrying 8 time slots for voice traffic. Determine the total number of simultaneous voice channels this deployment can offer.

Q4. During a site audit, engineers note the base station radiates 80 W while a nearby handset transmits just 4 W. They want to express this power gap as a decibel-based power control ratio (10·log10 of the ratio).

Q5. A drive test captures a serving signal at -60 dBm alongside co-channel interference at -92 dBm. The RF team needs the Carrier-to-Interference ratio (C/I) in dB to assess link quality.

Q6. An operator holds 42 MHz of spectrum, assigning 300 kHz per channel, and reuses frequencies across a 7-cell pattern. How many channels does a single cell end up with?

Q7. A transmitter delivers 64 W of power to a receiver. Engineers then move the receiver twice as far away, in an environment with a path-loss exponent of 4. What received power should they now expect?

Q8. Planners want to double a cell's radius while keeping the same SNR at the new edge, in an environment obeying an inverse-square power law. By what factor should transmit power be scaled up?

Q9. A network is built with a cluster size of 4, and every cell in the cluster is provisioned with 15 voice channels. What is the total channel count across the entire cluster?

Q10. A drive test records a vehicle traveling at 100 km/h while camped on a 900 MHz carrier. Estimate the Doppler shift the receiver experiences.

Q11. A TDMA frame lasting 5 ms is divided into 8 equal time slots. What is the duration of a single slot?

Q12. Spectrum planners have 24 MHz of allocated bandwidth and fix carrier spacing at 200 kHz for a GSM rollout. How many carrier frequencies can be created from this allocation?

Q13. A handset transmits at 8 W, but the signal experiences 33 dB of path loss on its way to the base station. Estimate the power the base station receiver actually sees, in dBm.

Q14. A network engineer uses 1 m as the reference distance while assessing a 1900 MHz deployment, where cell sites are spaced 2.5 km apart. Estimate the free-space path loss over that distance.

Q15. A cluster consists of 6 cells, each able to carry up to 250 simultaneous calls at peak. What is the cluster's total call-handling capacity?

Q16. A switching subsystem is rated to process 4000 new call requests every second, and calls last 3 minutes on average. How many calls could be active in the system at any one time?

Q17. Engineers assign 150 channels to a base station subsystem, each channel occupying 25 kHz. What total bandwidth does this base station consume?

Q18. Spectrum planners insert a 40 kHz guard band between each of 30 adjacent frequency bands to prevent interference. How much total bandwidth ends up wasted because of these guard bands?

Q19. A vehicle cruises at 90 km/h through a region of 1.5 km-radius cells. Using the cell diameter as the crossing distance, roughly how many handoffs would occur over an hour of travel?

Q20. A carrier's network hierarchy has 8 regional centers, each overseeing 40 base stations, with each base station serving 250 users. What is the network's total user-handling capacity?

Q21. Coverage planners want to blanket a 400 sq km region using hexagonal cells of 400 m radius (approximate cell area as R^2). Roughly how many cells will the rollout require?

Q22. An FDMA system allots 30 kHz to each channel, drawing from a total spectrum pool of 6 MHz. How many channels can be carved out of this allocation?

Q23. A market study notes that 1200 subscribers each place 4 calls per hour, averaging 2.5 minutes per call. What total offered traffic, in Erlangs, does this represent?

Q24. A 2.5 GHz deployment has 40 MHz of bandwidth, and each connected user consumes 400 kHz. What is the maximum number of users the band can serve at once?

Q25. A carrier's base stations each cap out at 600 simultaneous users, and the network comprises 1200 such stations. What total user capacity does the network offer?

Q26. A paging scheme sweeps through 4 cells at a time across a 200-cell network. In the worst case, how many paging rounds are needed to locate a subscriber?

Q27. A core network is engineered for 100,000 simultaneous connections. During a busy period it receives 40,000 call attempts per minute, each lasting 3 minutes on average. What fraction of the offered traffic ends up blocked?

Q28. A network policy lets 15% of its 600,000 subscribers roam onto partner networks each hour. At any given moment, how many subscribers are likely roaming?

Q29. A switching center is capable of processing 12,000 calls per second, but is currently handling a peak load of 9,000 calls per second. What percentage of its capacity is being utilized?

Q30. An operator reuses frequencies with a factor of 4 across a network holding 48 MHz of spectrum, with 200 kHz per channel. How many channels land in each cell?

Q31. A 900 MHz GSM deployment uses 200 kHz channel spacing across a 30 MHz allocation. How many channels can this spectrum support?

Q32. Over the course of an hour, a network logs 2500 handoff attempts, 99% of which succeed. How many calls were successfully handed off?

Q33. A mobile handset radiates 12 W, and the link experiences 45 dB of path loss before reaching the base station. What power level does the base station receiver detect?

Q34. A 7-cell cluster is deployed where each cell can serve up to 900 simultaneous users. What is the cluster's overall user capacity?

Q35. A TDMA carrier's frame spans 4.8 ms and is split into 8 equal slots. How long is each individual slot?

Q36. A mobile device moves at 150 km/h while connected to a 1.8 GHz carrier. What Doppler shift should the receiver expect?

Q37. A core switch is dimensioned for 250,000 simultaneous calls. It is fielding calls at a rate of 70,000 per minute, each averaging 4 minutes in duration. What share of the offered load gets blocked?

Q38. A hexagonal layout groups 9 cells into a cluster, and each cell can host 600 users. What is the cluster's combined user capacity?

Q39. A switching center processes 60,000 calls per second, with 0.4% of them dropped due to network issues. How many calls are being dropped every second?

Q40. A base station in an 1800 MHz network transmits at 43 dBm. At the edge of its 2 km-radius cell, the received signal measures -87 dBm. What path loss has the signal undergone?

Q41. A single base station covers 2.5 km² of territory, and the operator plans to deploy 150 identical stations. What total geographic area will the rollout cover?

Q42. An operator applies a frequency reuse factor of 6 over a 48 MHz spectrum allocation, with each channel taking up 200 kHz. How many channels does each cell get?

Q43. A paging system processes 6000 users per cycle, and the subscriber base totals 150,000 users. How many paging cycles are needed to sweep the entire base?

Q44. A phone call runs for 9 minutes, and during that time the network hands the call off roughly every 3 minutes. How many handoffs took place over the call?

Q45. A receiver picks up a wanted signal at -85 dBm against a noise floor of -115 dBm. What signal-to-noise ratio, in dB, does this represent?

Q46. A 4G cell is serving 250 users simultaneously, each needing a guaranteed 150 kbps. What is the minimum bandwidth the cell must provide to support them all?

Q47. A base station handles 1500 subscribers, each averaging 4 calls per hour with a 2.5-minute call length. What total traffic, in Erlangs, does this generate?

Q48. A network reserves 250 separate control channels, each 12 kHz wide, for signaling. What total bandwidth is tied up in control channels?

Q49. A network is carrying 1500 simultaneous connections, each streaming at 600 kbps on average. What total throughput must the network support?

Q50. A telecom operator wants each tower to cover a 4 km radius (treating coverage as a circle, area = πR^2) and needs to blanket a 350 km^2 service region. Roughly how many towers will the rollout require?

Rubrics for Calculation

Soln 1 Total spectrum = 45 MHz = 45,000 KHz. Full-duplex channel needs 2 simplex channels = 2 × 25 = 50 KHz. Channels = 45,000/50 = 900

Soln 2 Channels per cell = Total Channels / N = 560/7 = 80

Soln 3 Total voice channels = Carriers × slots/carries
= 100 × 8 = 800

Soln 4 Power ratio = 80/4 = 20
Ratio in dB = $10 \times \log_{10}(20)$ = 13.01 dB ≈ 13 dB

Soln 5 C/I (dB) = Signal (dBm) − Interference (dBm)
= −60 − (−92) = 32 dB

Soln 6 Total channels = 42,000 KHz / 300 KHz = 140
Channels per cell = 140/7 = 20

Soln 7 Received power $1/d^n$
Doubling distance with n=4 reduces power by $2^4 = 16\times$, New power = $\frac{64}{16} = 4W$

Soln 8 With n=2 (inverse square law), doubling radius requires power increases by 2^2 = 4x to maintain same edge SNR

Soln 9 Totals channels in cluster = cluster size × channels / cell
= 4 × 15 = 60

Soln 10 v = 100 Km/hr = 27.78 m/s
Doppler shift = $v \times \frac{f}{c}$ = $27.78 \times 900 \times 10^6 / 3 \times 10^8$ ≈ 83.3 Hz ≈ 83 Hz

Soln 11 Slot duration = $\frac{\text{Frame time}}{\text{Slots}} = \frac{5}{8}$ = 0.625 ms

Soln 12 Carriers = $\frac{\text{Total spectrum}}{\text{Spacing}} = \frac{24000 \text{ KHz}}{200 \text{ KHz}}$ = 120

Soln 13 Tx power in dBm = $10 \log_{10} (8000 \, mW) \approx 39.03 \, dBm$

Received power = $39.03 - 33 = 6.03 \, dBm \approx 6 \, dBm$

Soln 14 Using the log-distance path loss model referenced to 1m, scaling the distance to 2.5 Km gives an approximate free-space path loss of $\approx 123 \, dB$

Soln 15 Total capacity = $\frac{\text{Calls}}{\text{Cell} \times \text{cells}} = 250 \times 6 = 1500$

Soln 16 Active calls supported = arrival rate × avg duration

$\approx$ 4000 calls / sec × 180 sec = 720,000

Soln 17 Total bandwidth = channels × bandwidth

$= 150 \times 25 \, KHz = 3750 \, KHz = 3.75 \, MHz$

Soln 18 Number of guard bands between 30 bands = 29

Wasted bandwidth $= 29 \times 40 \, KHz = 1160 \, KHz = 1.16 \, MHz$

Soln 19 Speed = 90 Km/h = 25 m/s. Cell diameter = $2 \times 1.5 \, Km = 3000 \, m$

Time per cell = $\frac{3000}{25} = 120 \, s$ $\frac{\text{Handoffs}}{\text{hour}} = \frac{3600}{120} = 30$

Soln 20 Total users = regional centers × base stations × users

$= 8 \times 40 \times 250 = 80{,}000$

Soln 21 Approximate cell area $\approx R^2 = (0.4 \, Km)^2 = 0.16 \, Km^2$

Number of cells = $\frac{400}{0.16} = 2500$

Soln 22 Channels = $\frac{\text{Total spectrum}}{\text{Channel width}} = \frac{6000 \, KHz}{30 \, KHz} = 200$

Soln 24 Users = $\frac{\text{Total bandwidth}}{\text{bandwidth per user}} = \frac{40000 \, KHz}{400 \, KHz} = 100$

Soln 25 : Total capacity = users × cells

$= 600 \times 1200 = 720{,}000$

Soln 23 : Total call-minutes = $1200 \times 4 \times 2.5 = 12000 \, \text{min/hr}$

Traffic in Erlangs = $\frac{12000}{60} = 200$

~~Soln 26 Total capacity = user/cells x cells~~

~~= 600 x 12.00 = 7,20,000~~

<u>Soln 26</u> Paging operation = $\frac{\text{Total cells}}{\text{cells per operation}} = \frac{200}{4} = 50$

<u>Soln 27</u> Offered traffic = arrival rate x avg duration = 40000 x 3 = 12000 Erlangs

Excess over capacity = 120000 - 100000 = 20000

Fraction blocked = 20000 / 120000 ≈ 0.17

<u>Soln 28</u> Roaming users = 15 % x 600000 = 90000

<u>Soln 29</u> Utilization = $\frac{\text{Peak traffic}}{\text{Processing rate}} = \frac{9000}{12000} = 0.75 = 75\%$

<u>Soln 30</u> Total channels = $\frac{\text{Bandwidth}}{\text{channel width}} = \frac{48000}{200} = 240$

Channels per cell = $\frac{240}{4} = 60$

<u>Soln 31</u> channels = $\frac{\text{Total spectrum}}{\text{Channel width}} = \frac{30000}{200} = 150$

<u>Soln 32</u> Successful handoffs = 99 % x 2500 = 2475

<u>Soln 33</u> Tx power in dBm = 10 log10 (12000 mW) ≈ 40.79 dBm

Recieved power = 40.79 - 45 ≈ -4.21 dBm ≈ -4 dBm

<u>Soln 34</u> Total capacity = users / cell x cells = 900 x 7 = 6300

<u>Soln 35</u> Slot duration = $\frac{\text{Frame time}}{\text{slots}} = \frac{4.8}{8} = 0.6$ ms

<u>Soln 36</u> v = 150 km/h = 41.67 m/s   Doppler shift = $v \times \frac{f}{c} = \frac{41.67 \times 1.8 \times 10^9}{3 \times 10^8} = 250$ Hz

<u>Soln 37</u> offered traffic = 70000 x 4 = 2,80,000 Erlangs

Excess = 280000 - 250000 = 30000   Fraction blocked = $\frac{30000}{280000} \approx 0.11$

<u>Soln 38</u> Total users in cluster = users / cell x cells in cluster

= 600 x 9 = 5400

<u>Soln 39</u> Calls dropped / sec = 0.4 % x 60000 = 240

<u>Soln 40</u> Path loss (dB) = Tx Power (dBm) - Rx Power (dBm)

= 43 - (-87) = 130 dB

Soln 41 Total coverage = station × no. of stations
$= 2.5 \times 150 = 375\ Km^2$

Soln 42 Total channels $= \frac{48000}{200} = 240$

channels per cell $= \frac{240}{6} = 40$

Soln 43 Paging cycles $= \frac{\text{Total users}}{\text{user per cycle}} = \frac{150000}{6000} = 25$

Soln 44 Handoffs per call $= \frac{\text{call duration}}{\text{handoff interval}} = \frac{9}{3} = 3$

Soln 45 SNR (dB) = Signal (dBm) - Noise (dBm) $= -85 - (-115) = 30dB$

Soln 46 Minimum bandwidth = users × rate
$= 250 \times 150\ Kbps$
$= 37500\ Kbps = 37.5\ Mbps$

Soln 47 Total call-minutes / hr = users × calls × avg duration
$= 1500 \times 4 \times 2.5 = 15000\ min$
$= \frac{15000}{60} = 250$

Soln 48 Total bandwidth = Channels × bandwidth
$= 250 \times 12 = 3000\ KHz = 3\ MHz$

Soln 49 Total throughput = connections × rate
$= 1500 \times 600 = 900,000\ Kbps$
$= 900\ Mbps$

Soln 50 Area per tower $= \pi R^2$
$= \pi \times 4^2 = 50.27\ km^2$

Towers needed $= \frac{350}{50.27} = 6.96 = 7$